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DRIVE SYSTEM INSTRUCTIONS

DISCONNECTABLE RECTIFIER BRIDGES For SCR EXCITATION SYSTEMS

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

These instructions are to be used as a guide for installing, operating and maintaining the options on the SCR excitation system supplied on MS5001, MS6001 and MS7001 gas turbine driven generator.

Only personnel that are adequately trained and are thoroughly familiar with the equipment and these instructions should install, operate and maintain this equipment.

These instructions are to be used in conjunction with the excitation system instructions and installation instructions.

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* CAUTION *
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THERE IS A POSSIBILITY OF CONTROL MALFUNCTION IF A RADIO TRANSMITTER IS KEYED IN THE VICINITY OF PROCESS CONTROL OR POWER CONVERSION EQUIPMENT. THE INTERFERENCE MAY BE ESPECIALLY NOTICEABLE WITHIN FOUR FEET OF EQUIPMENT WITH PANEL DOORS OPEN.

It is recommended that walkie talkie type radios not be used in close proximity of operating electronic equipment.

Keeping panel doors closed and having proper frame to ground grid system connections will minimize the problem but may not eliminate it.

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* WARNING *
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THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

Throughout this instruction book there are two conventions used for highlighting names. Abbreviations, signal names, names of adjustable devices and test points are capitalized as in the following examples: PT, VREF, 2P (TEST), 6TP. However, some device names also capitalized, such as 59EX, 1T, 1VR, etc. The names of modules, submodules and printed wiring boards (cards) are underlined when they occur within the body of the text. Examples of this type of highlight are as follows: Signal Conditioner module, Auto Regulator PWB, AC/DC PWB.

RECEIVING AND STORAGE

As soon as the equipment is received, it should be examined for any damage that might have been sustained in transit and all shipping material removed. If injury or rough handling is evident, a damage claim shall be filed immediately with the carrier and the nearest General Electric sales office should be notified promptly.

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If the equipment is to be stored the space heaters must be energized to prevent moisture condensation.

ABBREVIATIONS

The following abbreviations are used in this instruction book.

AC, ac Alternating Current

AFFL Amps, Field, Full Load

AFNL Amps, Field, No Load

ARCC Active and Reactive Current Compensation

CT Current Transformer

D/A Digital to Analog

DC, dc Direct Current

GAC Generator Auxiliary Compartment

GEC Generator Excitation Compartment

I Current

IC Integrated Circuit

LDC Line Drop Compensator

LED Light Emitting Diode

LSB Least Significant Bit

MAX Maximum

MIN Minimum

MW Megawatts

PCB Printed Circuit Board (same as PWB)

PB Pushbutton

PF, pf Power Factor

PPT Power Potential Transformer

PS Power Supply

PSS Power System Stabilizer

PT Potential Transformer

PU Per Unit

PWB Printed Wiring Board (same as PCB)

RC Resistor Capacitor

RCC Reactive Current Compensation

SCR Silicon Controlled Rectifier

SVA Static Voltage Adjuster

TP, Test Point

TVM Transfer Voltmeter

URAL Under-excited Reactive Ampere Limit

V Volts

VARS Volt Amperes, Reactive

VFFL Volts, Field, Full Load

VFNL Volts, Field, No Load

V/Hz Volts/Hertz

INTRODUCTION

The Disconnectable Rectifier Bridge option for static excitation systems on gas turbines provides for one rectifier bridge to be disconnected from the field circuits during normal operation. This allows for maintenance and repair work to be done on the disconnected bridge without removing the unit from service.

In general, several rectifier bridges, connected in parallel, may be supplied with an excitation system in order to provide the rated field current to the generator. The number supplied is normally the minimum needed to meet the maximum field current requirement, and these bridges are permanently connected together. The Disconnectable Rectifier Bridge option, on the other hand, provides disconnects on the input and output of each bridge so it can be isolated from the excitation system power circuits. In addition, an extra bridge is supplied so the unit may operate at full load with one bridge out of service. Figure 1 shows the general configuration when this option is added to a two-bridge standard system.

DESCRIPTION

The Disconnectable Rectifier Bridge option adds the following material to the standard excitation system:

1) One additional rectifier bridge assembly and connecting buswork,

2) One disconnect switch for each bridge, and

3) One status indicating LED assembly for each bridge.

The bridge disconnect is a five-pole switch that, when opened, removes power from the three AC input lines and disconnects the two DC output lines.

Each LED assembly consists of one green and one red LED for each main rectifier in the bridge. No LEDs are supplied for the series and free-wheeling rectifiers on Potential Source Excitation Systems. The assembly also contains a current limiting resistor for each pair of LEDs.

PRINCIPLES OF OPERATION

The Disconnectable Rectifier Bridge option is designed to allow the performance of repairs and maintenance on a rectifier bridge in a static excitation system without any interruption of service. Consider the system in Figure 1 operating under full load conditions. Should maintenance be required on bridge 1, for example, the disconnect may be opened to remove that bridge from service. Bridges 2 and 3 immediately assume the current load previously carried by bridge 1. Since the system has one redundant bridge, there is no danger of exceeding the bridge current rating when operating under these conditions. Opening the disconnect switch does not interrupt the current flowing to the field. It only forces the other bridge(s) to carry more current.

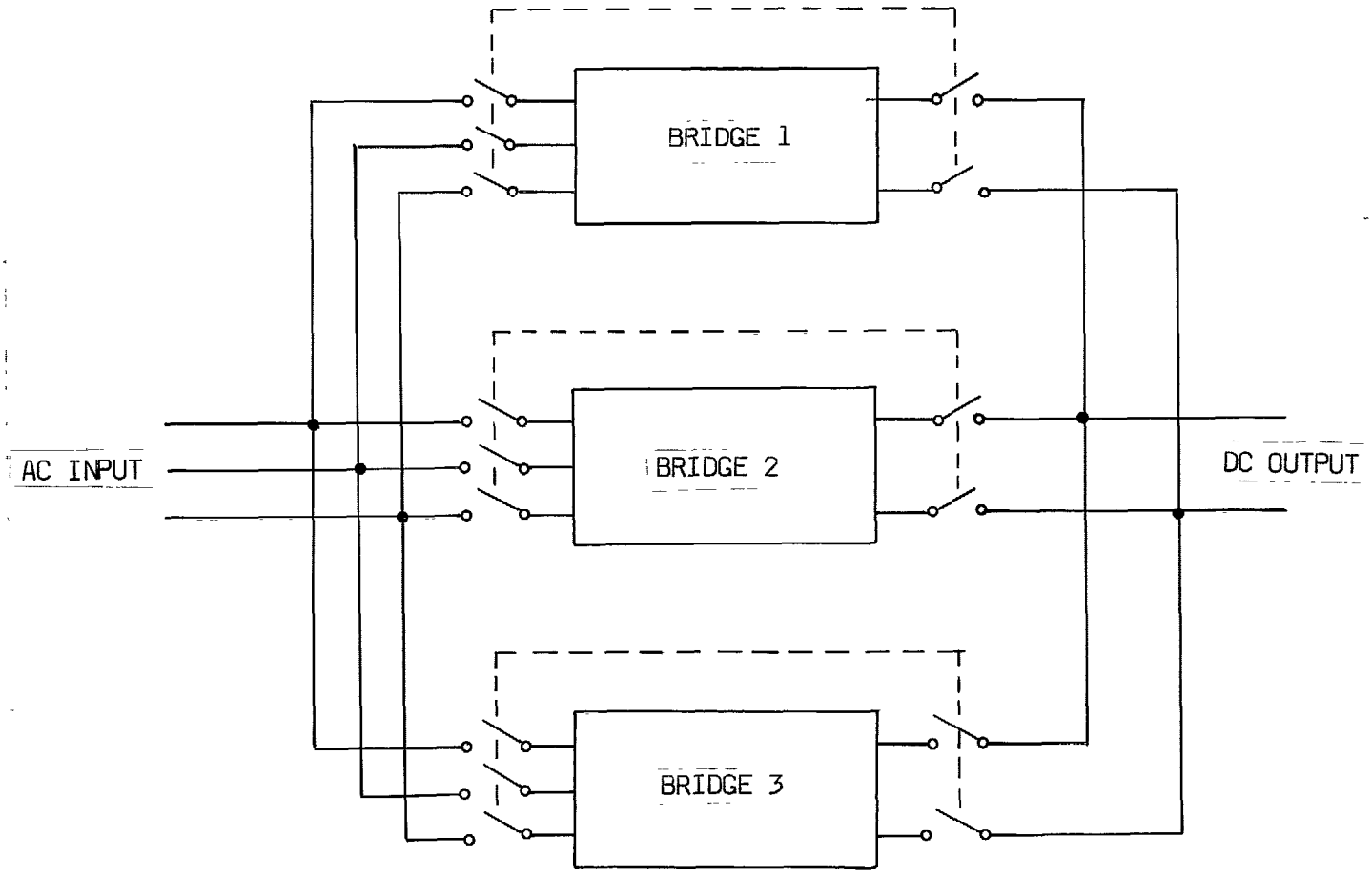


Figure 1.
Rectifier Circuit With Disconnects

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* CAUTION *
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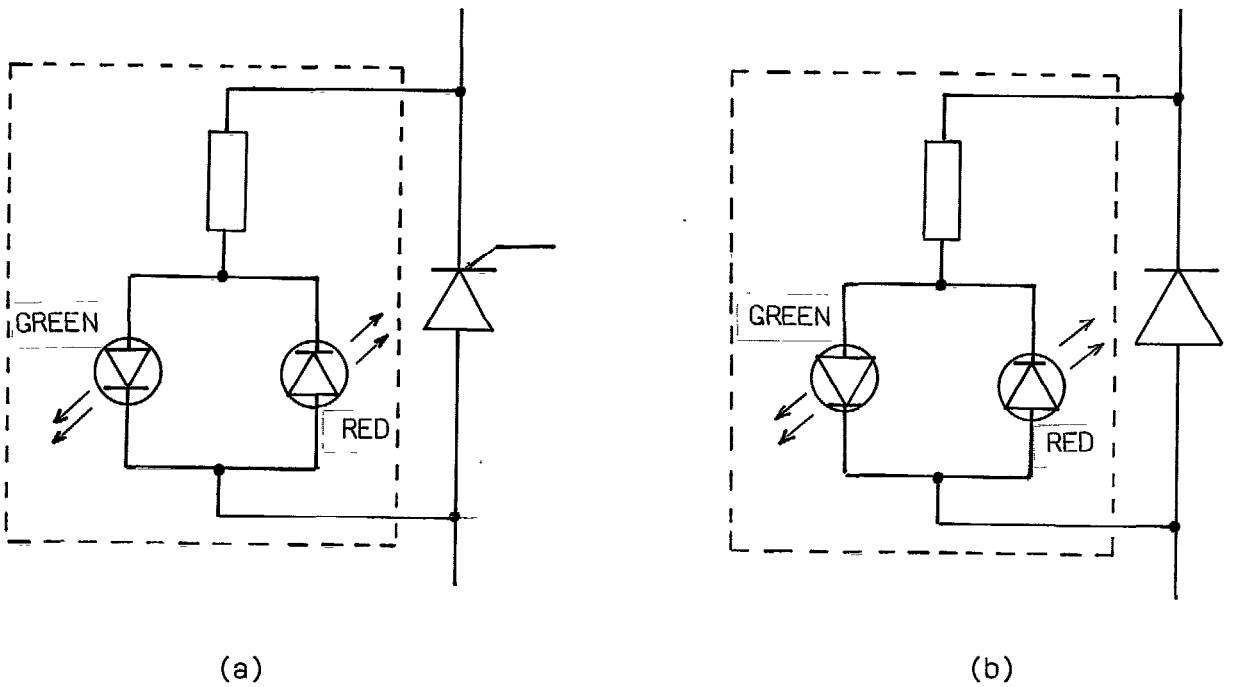
DISCONNECTING MORE THAN ONE RECTIFIER BRIDGE AT A TIME MAY RESULT IN OVERLOADING THE BRIDGE(S) REMAINING IN SERVICE. OPERATE ONLY IN THE DC MODE AND DO NOT EXCEED THE RATING OF THE BRIDGE(S) IN SERVICE.

When the maintenance on bridge 1 is complete, it may be returned to service by reclosing the disconnect switch. The three parallel bridges will again share the current load. Bridge 2 or bridge 3 may also be removed from service and restored in the manner described above.

The status indicating LEDs supplied with this option provide a visual display of the relative condition of each power rectifier in the bridges. Each SCR has two LEDs connected between its anode and cathode (See Figure 2). Normally the two LEDs should glow with unequal brightness. When the firing angle of the SCR is advanced, the green LED will be bright; when retarded, the r d LED will be the brighter. Should an SCR short and/or its fuse blow, the associated lights will go out. If an SCR fails to fire (and no parallel path exists), both LEDs will be bright. When a bridge is removed from service, all of its associated LEDs will go out.

NOTE

With more than one bridge in service, a non-firing SCR will not be detectable by its LED indicators since its corresponding SCR(s) in the other bridge(s) will be conducting.



(a) (b)

Figure 2.

(a) LED Circuit on an SCR (b) LED Circuit on a Diode

Each power diode also has two LEDs connected between its anode and cathode (See Figure 2). The normal condition is the green LED on and the red LED off. Like the SCR, a shorted diode results in both LEDs going out, and if a diode is not conducting, both LED's will be on with equal intensity.

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INITIAL CHECKS

1. Verify that the disconnect switches are connected according to the elementary diagram for the AC power circuit.

2. Wire check each LED assembly (see elementary sheet for rectifier bridge circuit).

3. Close all disconnect switches.

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* CAUTION *
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DO NOT HI-POT OR MEGGER ANY CIRCUITS AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE EXCITATION SYSTEM.

ON LINE/OFF LINE TESTS

The Disconnectable Rectifier Bridge option may be checked for proper operation while operating either on line or off line according to the following procedure:

1. With all bridges in service (disconnects closed), verify that the LED indicators are illuminated according to Table 1.

2. While monitoring field voltage and current, open one disconnect switch. There should be no change in excitation. All LEDs for the disconnected bridge should go out. Reclose the disconnect switch.

Table 1

Excitation System	SCRs				Diodes	
	No Load		Full Load			
	Red	Green	Red	Green	Red	Green
Potential Source	Bright	Dim	Dim	Bright	Off	On
Shunt SCR	Dim	Bright	Bright	Dim	Off	On

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* CAUTION *
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DISCONNECTING MORE THAN ONE RECTIFIER BRIDGE AT A TIME MAY RESULT IN OVERLOADING THE BRIDGE(S) REMAINING IN SERVICE. OPERATE ONLY IN THE DC REGULATOR MODE AND DO NOT EXCEED THE RATING OF THE BRIDGE(S).

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* WARNING *
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OPENING A DISCONNECT SWITCH DOES NOT REMOVE POWER FROM THE GATE DRIVER CIRCUITS. ALL SAFETY PRECAUTIONS RELATIVE TO WORKING AROUND ENERGIZED EQUIPMENT SHOULD BE OBSERVED.

3. Repeat part 2 for each disconnect switch.

This completes the tests for the Disconnectable Rectifier Bridge option.

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